Introduction to Artificial Intelligence (INTRAI)

Module 2: How AI Systems Work

Data, Model and Prediction

Lesson Notes

KMVTECHUG

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The Simple Pipeline

Most modern AI follows a simple pipeline: data is used to build or adjust a model, and the model produces a prediction or decision when given new input.

Data might be photos of maize leaves, past mobile-money transactions, or text from support chats. The model is the structure that has learned patterns from that data. Prediction is the output: disease likely, transfer unusual, or suggested reply.

Remember three words: data, model, prediction.

Inputs and Goals

Every useful AI system has a goal: classify an image, rank a risk score, suggest the next word, translate a sentence.

Inputs must match what the model was built for. A model trained on English support tickets will struggle on handwritten notes in another language it has never seen. Clear goals and matching inputs matter more than flashy demos.

Rules vs Learning Systems

Rules-based systems follow explicit if-then instructions written by people. They are transparent but brittle when the world changes.

Learning systems adjust internal parameters from examples. They can handle messy patterns but depend heavily on data quality and can be harder to explain.

Many real products mix both. A fraud system may use learned scores plus hard rules for known scam patterns.

> ***Reflect:*** *For a spam filter, what is the data, what is the model doing, and what is the prediction?*

Closing

Module 2 continues with everyday examples and why data quality decides whether the pipeline works in real life.